

Working Efficiently with a Scientific Calculator

Desmos Scientific Calculator · Desmos

Senior Secondary (Years 11-12)

~30 minutes

Mathematics · Essential
Mathematics

NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://www.desmos.com/scientific>

Curriculum links: number · measurement · probability · statistics

Learning intentions

- I can use a scientific calculator efficiently for multi-step number and measurement calculations.
- I can use a calculator to compute basic probability and simple statistical measures.
- I can judge when an approximate or exact calculator result is appropriate for a given context.

Getting started

Open the Desmos Scientific Calculator. This worksheet uses it as your working calculator for a mixed set of number, measurement and probability problems, with no graphing required. Keep it open in a separate tab as you work through each task.

Tasks

1. A rectangular room measures 4.6 m by 3.85 m. Calculate its area and perimeter, rounding sensibly for a real-world measurement.

Working:

2. Calculate the volume of a cylindrical water tank with radius 1.2 m and height 2.5 m, using the calculator's pi function rather than a rounded value of pi.

Working:

3. Compare the result from the previous task using pi rounded to 3.14 versus the calculator's built-in pi. Explain any difference in your answers.

4. A bag contains 4 red, 5 blue and 3 green marbles. Use the calculator to find the probability of drawing a blue marble, expressed as a fraction, decimal and percentage.

Working:

5. A class of 20 students sat a test. 8 students scored above the class mean. Use the calculator to express this as a percentage of the class.

Working:

6. Describe a situation from everyday life, outside this worksheet, where using an exact value would matter more than a rounded decimal approximation, and one where rounding is perfectly fine.

7. Calculate the compound value of \$2000 invested at 4% per annum for 3 years, using the calculator's power function.

Working:

Extension (optional)

Use the calculator's statistics mode, or work through the formula step by step, to find the standard deviation of the dataset 12, 15, 18, 21, 24. Explain what this value tells you about the spread of the data.
